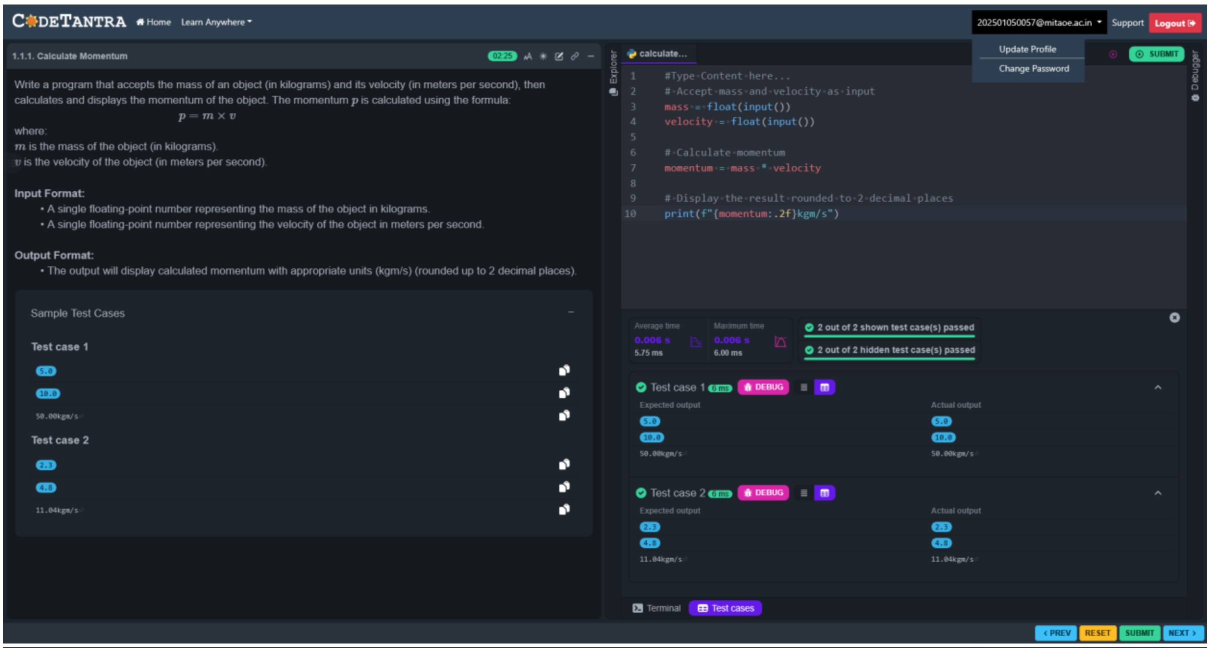
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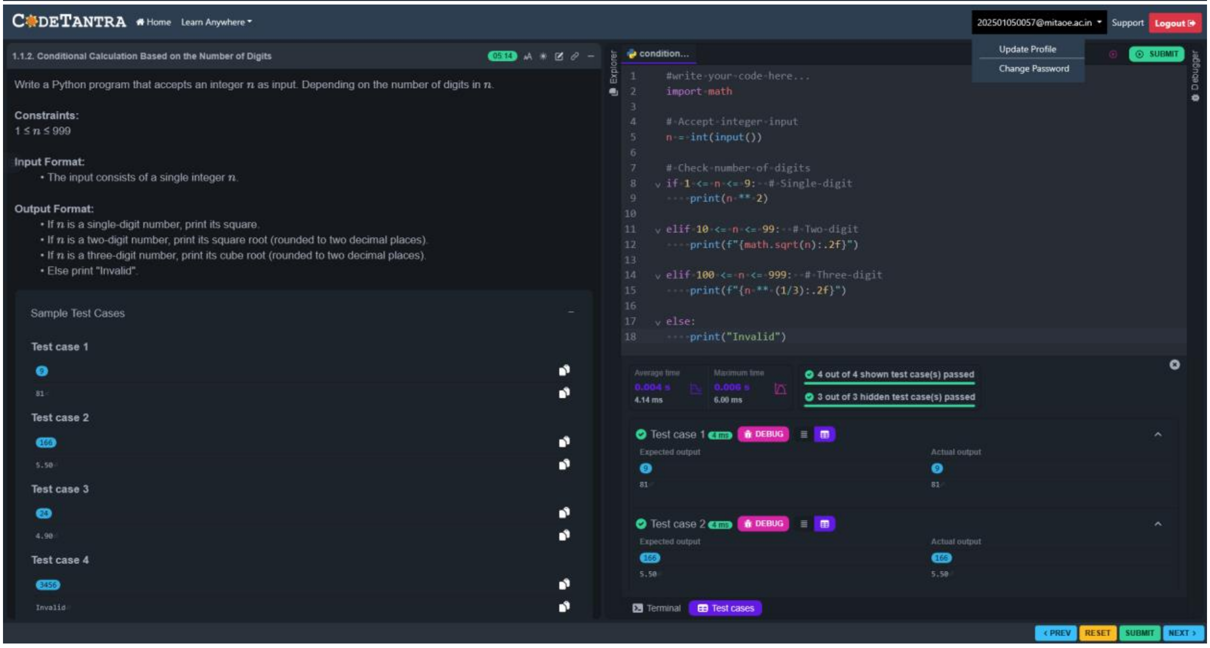
1.1.1. Calculate Momentum
Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:
p = m × v
where:
m is the mass of the object (in kilograms).
v is the velocity of the object (in meters per second).
Input Format:
• A single floating-point number representing the mass of the object in kilograms.
• A single floating-point number representing the velocity of the object in meters per second.
Output Format:
• The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).
Sample Test Cases
Test case 1
5.0
10.0
50.00kgm/s
Test case 2
2.3
4.8
11.04kgm/s
calculate...
SUBMIT
#Type Content here...
# Accept mass and velocity as input
mass = float(input())
velocity = float(input())
# Calculate momentum
momentum = mass * velocity
# Display the result rounded to 2 decimal places
print(f"{momentum:.2f}kgm/s")
Average time
0.006 s
5.75 ms
Maximum time
0.006 s
6.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
DEBUG
Expected output
Actual output
5.0
10.0
50.00kgm/s
Test case 2
DEBUG
Expected output
Actual output
2.3
4.8
11.04kgm/s
Terminal
Test cases
PREV
RESET
SUBMIT
NEXT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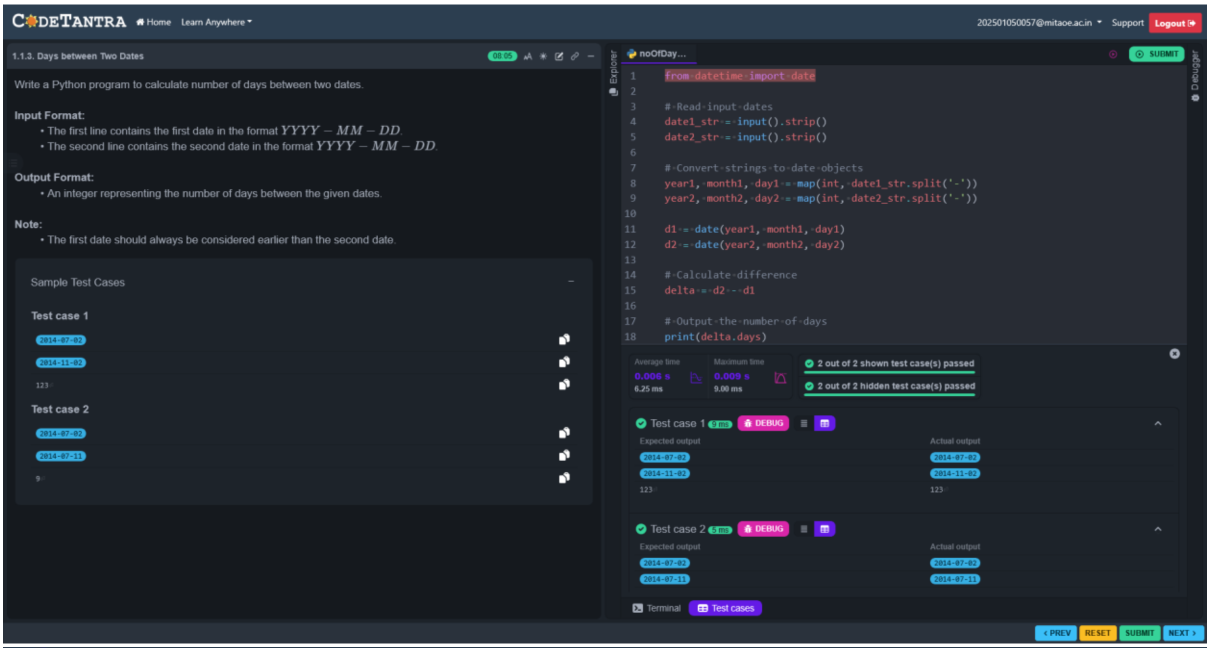
1.1.2. Conditional Calculation Based on the Number of Digits
Write a Python program that accepts an integer n as input. Depending on the number of digits in n.
Constraints:
1 ≤ n ≤ 999
Input Format:
• The input consists of a single integer n.
Output Format:
• If n is a single-digit number, print its square.
• If n is a two-digit number, print its square root (rounded to two decimal places).
• If n is a three-digit number, print its cube root (rounded to two decimal places).
• Else print "Invalid".
Sample Test Cases
Test case 1
9
81
Test case 2
166
5.50
Test case 3
34
4.90
Test case 4
3456
Invalid
condition...
SUBMIT
#write your code here...
import math
# Accept integer input
n = int(input())
# Check number of digits
if 1 <= n <= 9: # Single-digit
    print(n ** 2)
elif 10 <= n <= 99: # Two-digit
    print(f"{math.sqrt(n):.2f}")
elif 100 <= n <= 999: # Three-digit
    print(f"{n ** (1/3):.2f}")
else:
    print("Invalid")
Average time
0.004 s
4.14 ms
Maximum time
0.006 s
6.00 ms
4 out of 4 shown test case(s) passed
3 out of 3 hidden test case(s) passed
Test case 1
4 ms
DEBUG
Expected output
Actual output
9
81
Test case 2
4 ms
DEBUG
Expected output
Actual output
166
5.50
Terminal
Test cases
PREV
RESET
SUBMIT
NEXT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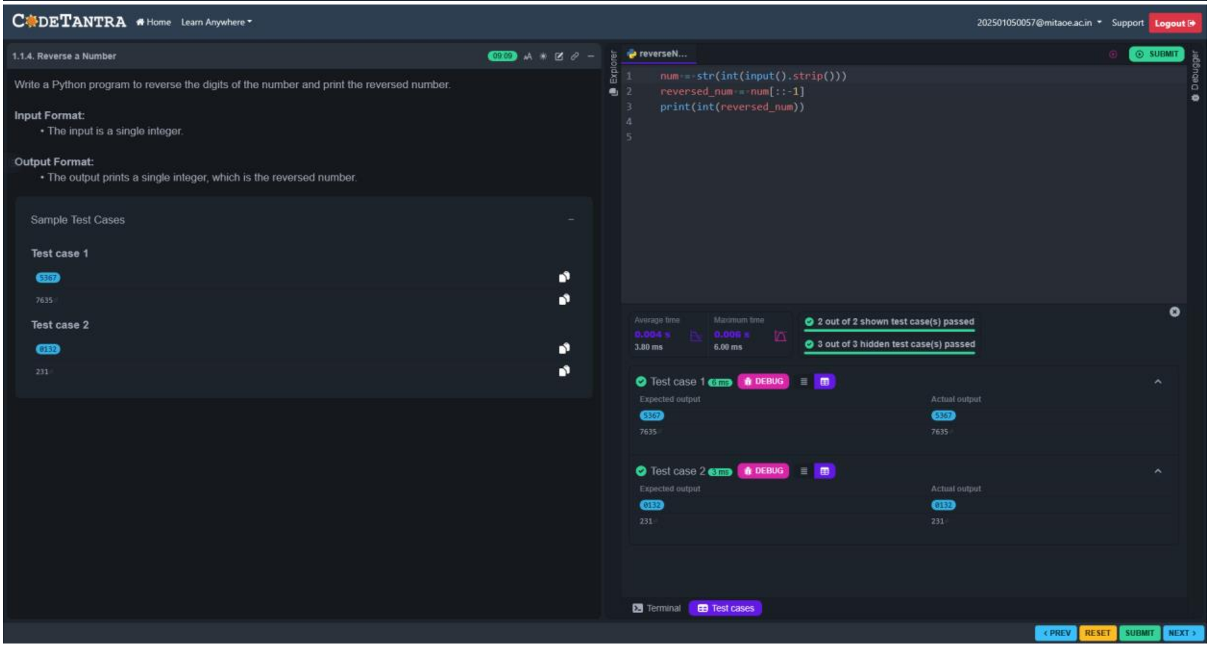
1.1.3. Days between Two Dates
Write a Python program to calculate number of days between two dates.
Input Format:
• The first line contains the first date in the format YYYY − MM − DD.
• The second line contains the second date in the format YYYY − MM − DD.
Output Format:
• An integer representing the number of days between the given dates.
Note:
• The first date should always be considered earlier than the second date.
Sample Test Cases
Test case 1
2014-07-02
2014-11-02
123
Test case 2
2014-07-02
2014-07-11
9
noOfDay...
SUBMIT
from datetime import date
# Read input dates
date1_str = input().strip()
date2_str = input().strip()
# Convert strings to date objects
year1, month1, day1 = map(int, date1_str.split('-'))
year2, month2, day2 = map(int, date2_str.split('-'))
d1 = date(year1, month1, day1)
d2 = date(year2, month2, day2)
# Calculate difference
delta = d2 - d1
# Output the number of days
print(delta.days)
Average time
0.006 s
6.25 ms
Maximum time
0.009 s
9.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
DEBUG
Expected output
2014-07-02
2014-11-02
123
Actual output
2014-07-02
2014-11-02
123
Test case 2
DEBUG
Expected output
2014-07-02
2014-07-11
Actual output
2014-07-02
2014-07-11
Terminal
Test cases
PREV
RESET
SUBMIT
NEXT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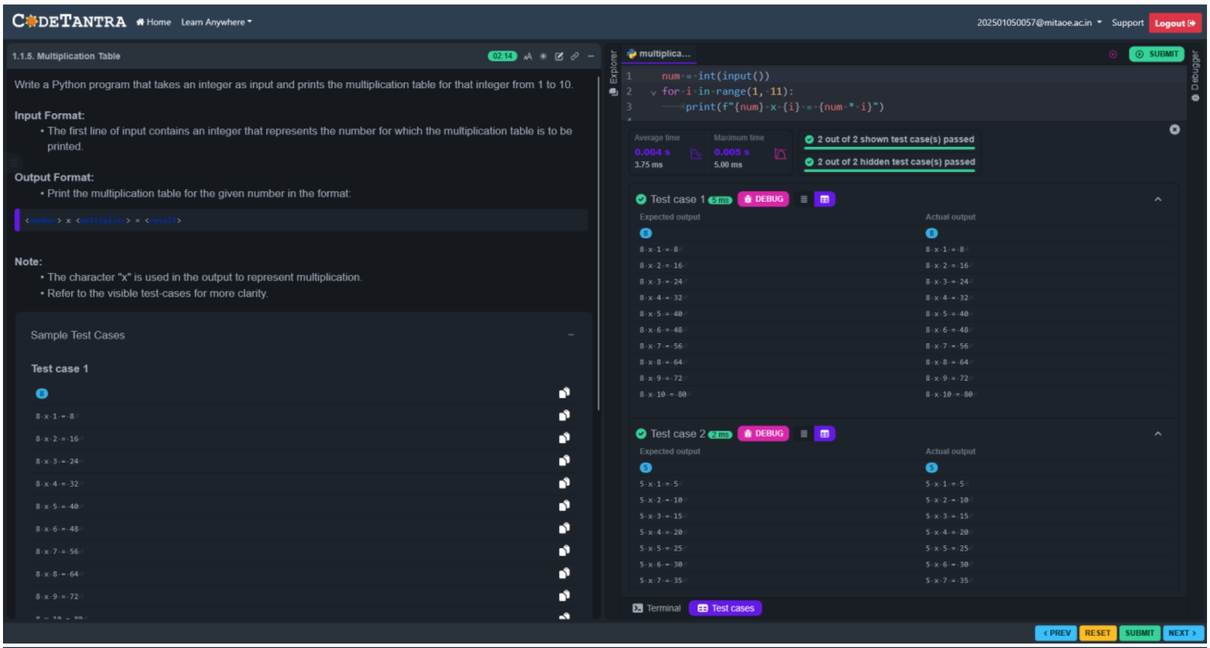
1.1.4. Reverse a Number
Write a Python program to reverse the digits of the number and print the reversed number.
Input Format:
• The input is a single integer.
Output Format:
• The output prints a single integer, which is the reversed number.
Sample Test Cases
Test case 1
5367
7635
Test case 2
0132
231
reverseN...
SUBMIT
num = str(int(input().strip()))
reversed_num = num[::-1]
print(int(reversed_num))
Average time
0.004 s
3.80 ms
Maximum time
0.006 s
6.00 ms
2 out of 2 shown test case(s) passed
3 out of 3 hidden test case(s) passed
Test case 1
DEBUG
Expected output
5367
7635
Actual output
5367
7635
Test case 2
DEBUG
Expected output
0132
231
Actual output
0132
231
Terminal
Test cases
PREV
RESET
SUBMIT
NEXT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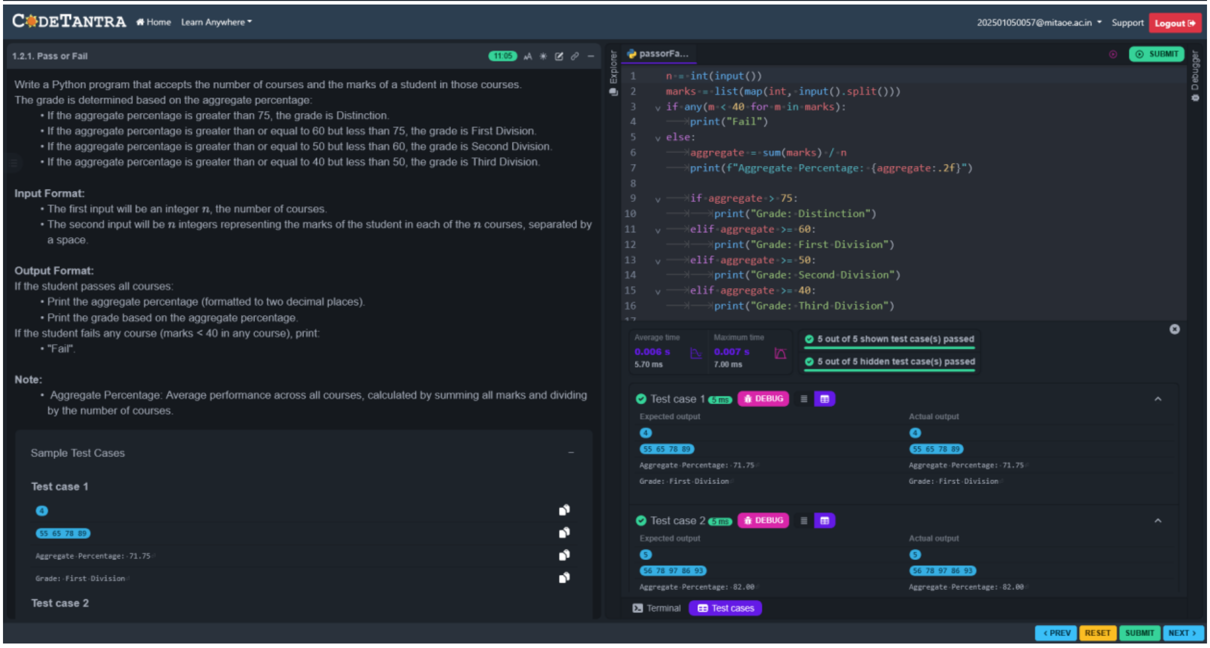
1.1.5. Multiplication Table
02:14
Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.
Input Format:
• The first line of input contains an integer that represents the number for which the multiplication table is to be printed.
Output Format:
• Print the multiplication table for the given number in the format:
<number> x <multiplier> = <result>
Note:
• The character "x" is used in the output to represent multiplication.
• Refer to the visible test-cases for more clarity.
Sample Test Cases
Test case 1
8
8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40
8 x 6 = 48
8 x 7 = 56
8 x 8 = 64
8 x 9 = 72
multiplica...
SUBMIT
num = int(input())
for i in range(1, 11):
print(f"{num} x {i} = {num * i}")
Average time
0.004 s
3.75 ms
Maximum time
0.005 s
5.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
5 ms
DEBUG
Expected output
Actual output
8
8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40
8 x 6 = 48
8 x 7 = 56
8 x 8 = 64
8 x 9 = 72
8 x 10 = 80
Test case 2
2 ms
DEBUG
Expected output
Actual output
5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
Terminal
Test cases
Explorer
Debugger
‹ PREV
RESET
SUBMIT
NEXT ›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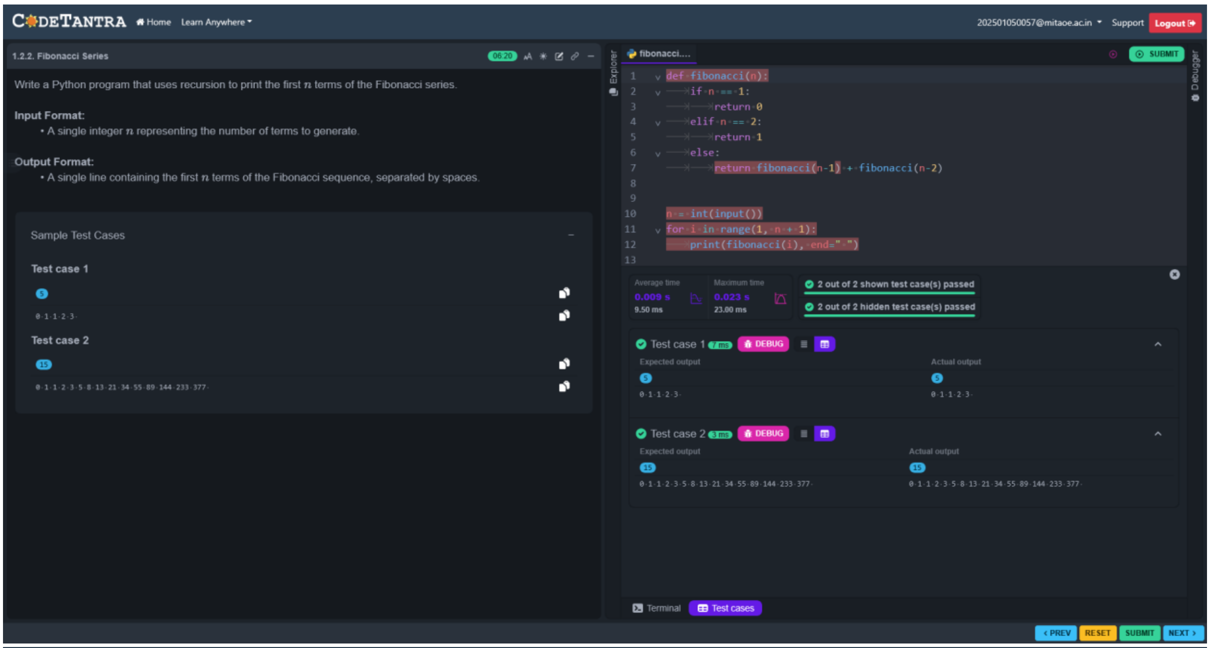
1.2.1. Pass or Fail
11:05
Write a Python program that accepts the number of courses and the marks of a student in those courses.
The grade is determined based on the aggregate percentage:
• If the aggregate percentage is greater than 75, the grade is Distinction.
• If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
• If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
• If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.
Input Format:
• The first input will be an integer n, the number of courses.
• The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.
Output Format:
If the student passes all courses:
• Print the aggregate percentage (formatted to two decimal places).
• Print the grade based on the aggregate percentage.
If the student fails any course (marks < 40 in any course), print:
• "Fail".
Note:
• Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.
Sample Test Cases
Test case 1
4
55 65 78 89
Aggregate Percentage: 71.75
Grade: First Division
Test case 2
passorFa...
SUBMIT
n = int(input())
marks = list(map(int, input().split()))
if any(m < 40 for m in marks):
print("Fail")
else:
aggregate = sum(marks) / n
print(f"Aggregate Percentage: {aggregate:.2f}")
if aggregate > 75:
print("Grade: Distinction")
elif aggregate >= 60:
print("Grade: First Division")
elif aggregate >= 50:
print("Grade: Second Division")
elif aggregate >= 40:
print("Grade: Third Division")
Average time
0.006 s
5.70 ms
Maximum time
0.007 s
7.00 ms
5 out of 5 shown test case(s) passed
5 out of 5 hidden test case(s) passed
Test case 1
5 ms
DEBUG
Expected output
Actual output
4
55 65 78 89
Aggregate Percentage: 71.75
Grade: First Division
Test case 2
5 ms
DEBUG
Expected output
Actual output
5
56 78 97 86 93
Aggregate Percentage: 82.00
Terminal
Test cases
Explorer
Debugger
‹ PREV
RESET
SUBMIT
NEXT ›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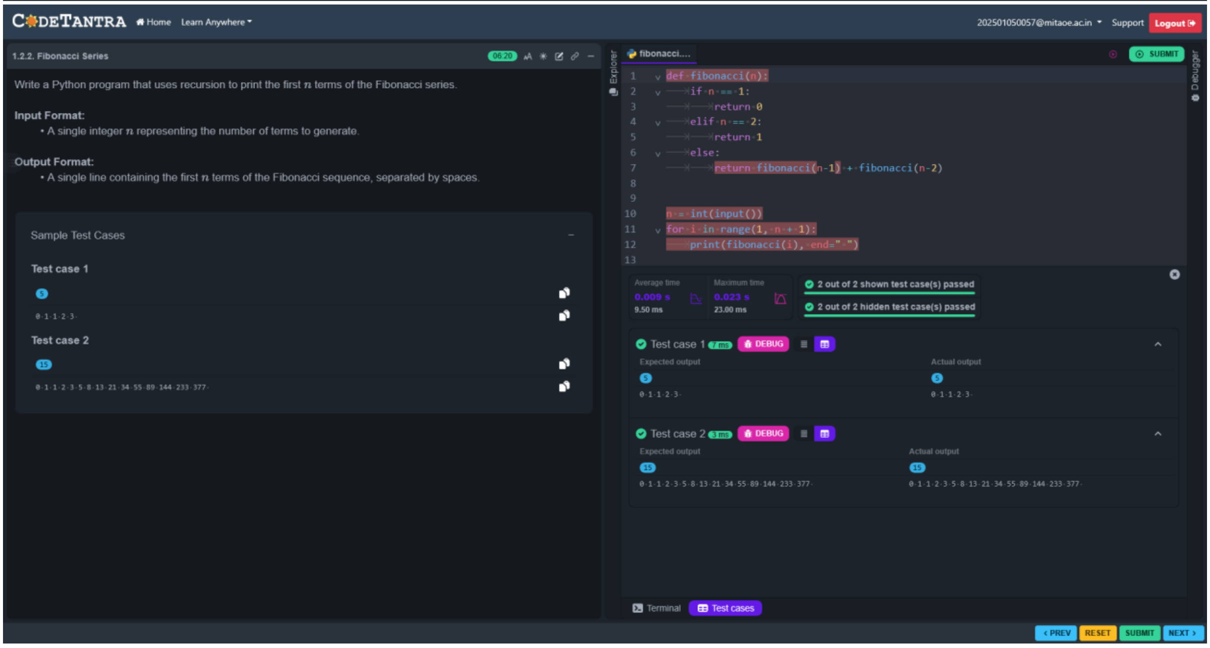
1.2.2. Fibonacci Series
Write a Python program that uses recursion to print the first n terms of the Fibonacci series.
Input Format:
• A single integer n representing the number of terms to generate.
Output Format:
• A single line containing the first n terms of the Fibonacci sequence, separated by spaces.
Sample Test Cases
Test case 1
5
0 1 1 2 3
Test case 2
15
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377
fibonacci....
SUBMIT
def fibonacci(n):
    if n == 1:
        return 0
    elif n == 2:
        return 1
    else:
        return fibonacci(n-1) + fibonacci(n-2)
n = int(input())
for i in range(1, n + 1):
    print(fibonacci(i), end=" ")
Average time
0.009 s
9.50 ms
Maximum time
0.023 s
23.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
7 ms
DEBUG
Expected output
5
0 1 1 2 3
Actual output
5
0 1 1 2 3
Test case 2
3 ms
DEBUG
Expected output
15
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377
Actual output
15
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377
Terminal
Test cases
PREV
RESET
SUBMIT
NEXT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1.2.2. Fibonacci Series
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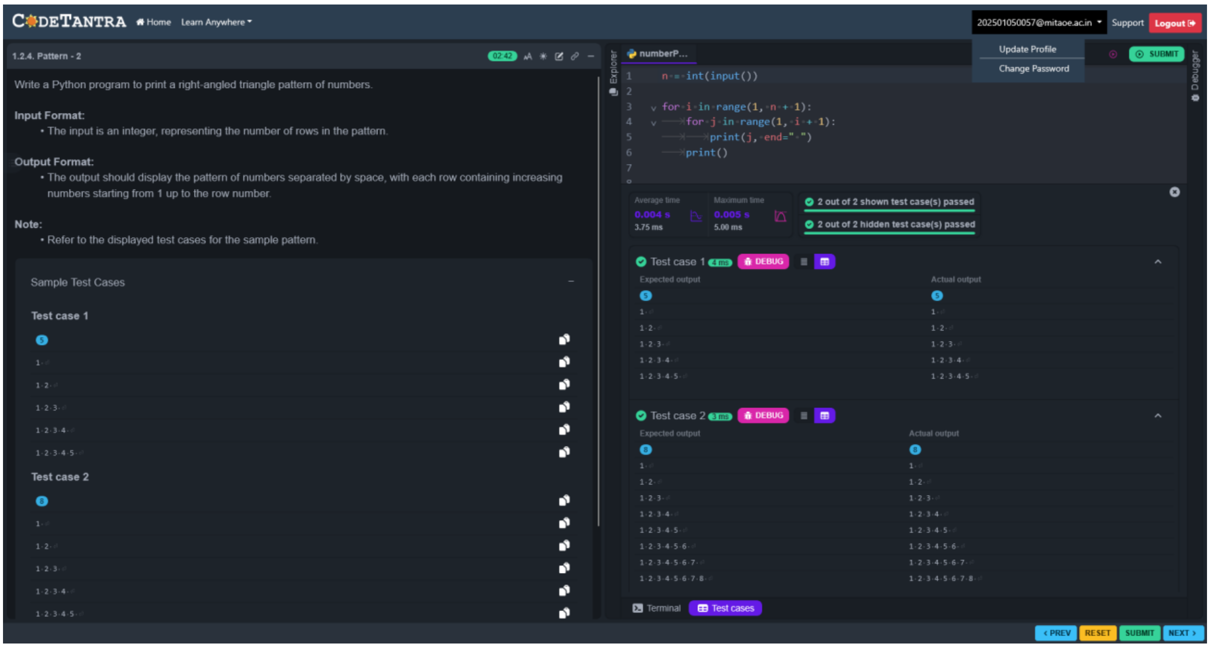
Input Format:
• A single integer n representing the number of terms to generate.
Output Format:
• A single line containing the first n terms of the Fibonacci sequence, separated by spaces.
Sample Test Cases
Test case 1
5
0 1 1 2 3
Test case 2
15
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377
fibonacci....
SUBMIT
def fibonacci(n):
    if n == 1:
        return 0
    elif n == 2:
        return 1
    else:
        return fibonacci(n-1) + fibonacci(n-2)
n = int(input())
for i in range(1, n + 1):
    print(fibonacci(i), end=" ")
Average time
0.009 s
9.50 ms
Maximum time
0.023 s
23.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
7 ms
DEBUG
Expected output
5
0 1 1 2 3
Actual output
5
0 1 1 2 3
Test case 2
3 ms
DEBUG
Expected output
15
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377
Actual output
15
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377
Terminal
Test cases
PREV
RESET
SUBMIT
NEXT



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1.2.4. Pattern - 2
Write a Python program to print a right-angled triangle pattern of numbers.
Input Format:
• The input is an integer, representing the number of rows in the pattern.
Output Format:
• The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.
Note:
• Refer to the displayed test cases for the sample pattern.
Sample Test Cases
Test case 1
5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
Test case 2
8
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
numberP...
SUBMIT
n = int(input())
for i in range(1, n + 1):
for j in range(1, i + 1):
print(j, end=" ")
print()
Average time
0.004 s
3.75 ms
Maximum time
0.005 s
5.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
4 ms
DEBUG
Expected output
Actual output
Test case 2
3 ms
DEBUG
Terminal
Test cases
PREV
RESET
SUBMIT
NEXT